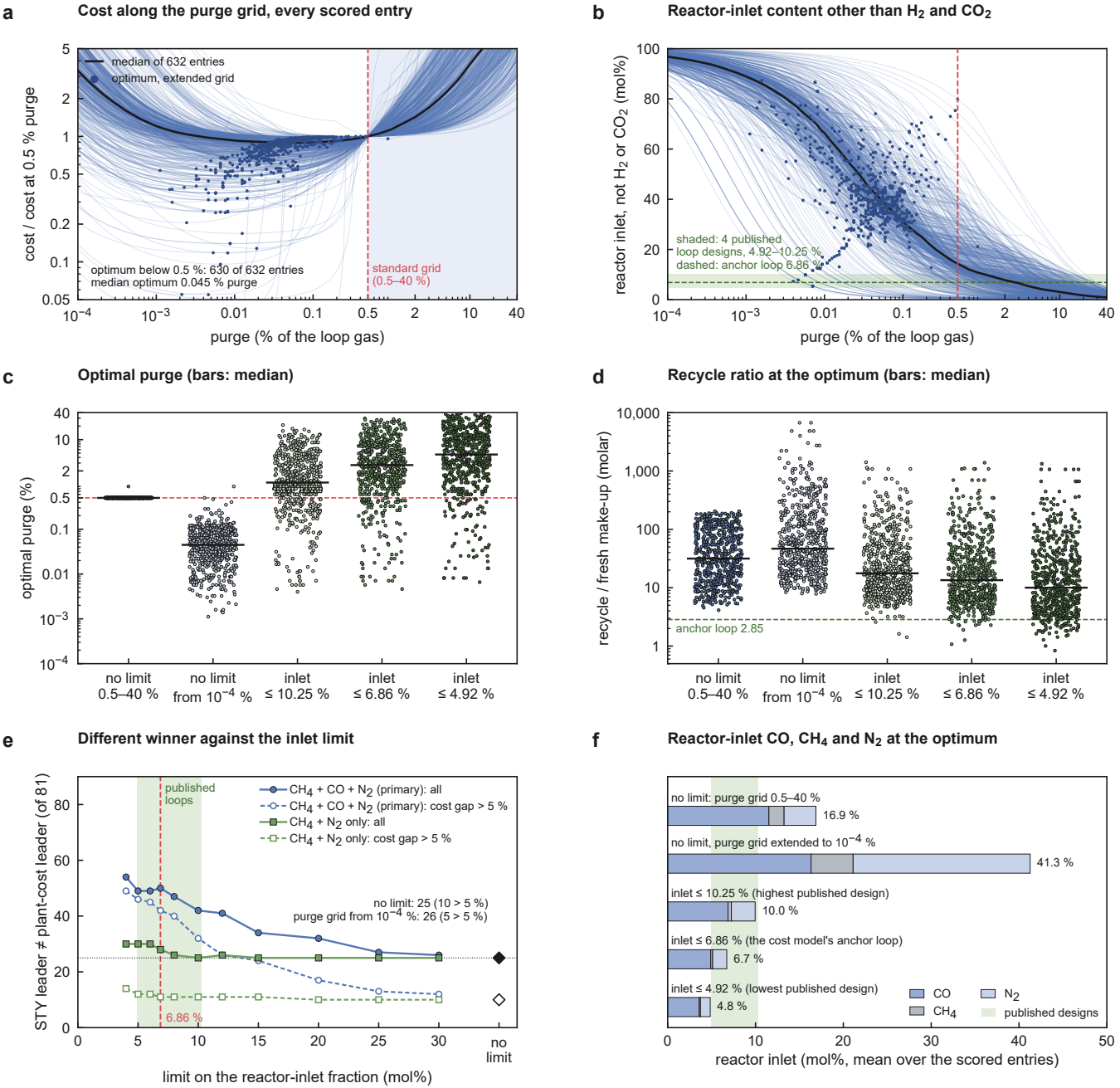

a
Cost along the purge grid, every scored entry
median of 632 entries
optimum, extended grid
optimum below 0.5 %: 630 of 632 entries
median optimum 0.045 % purge
standard grid
(0.5–40 %)
cost / cost at 0.5 % purge
purge (% of the loop gas)
b
Reactor-inlet content other than H2 and CO2
shaded: 4 published
loop designs, 4.92–10.25 %
dashed: anchor loop 6.86 %
reactor inlet, not H2 or CO2 (mol%)
purge (% of the loop gas)
c
Optimal purge (bars: median)
optimal purge (%)
no limit
0.5–40 %
no limit
from 10−4 %
inlet
≤ 10.25 %
inlet
≤ 6.86 %
inlet
≤ 4.92 %
d
Recycle ratio at the optimum (bars: median)
recycle / fresh make-up (molar)
anchor loop 2.85
e
Different winner against the inlet limit
published
loops
CH4 + CO + N2 (primary): all
CH4 + CO + N2 (primary): cost gap > 5 %
CH4 + N2 only: all
CH4 + N2 only: cost gap > 5 %
no limit: 25 (10 > 5 %)
purge grid from 10−4 %: 26 (5 > 5 %)
6.86 %
STY leader ≠ plant-cost leader (of 81)
limit on the reactor-inlet fraction (mol%)
f
Reactor-inlet CO, CH4 and N2 at the optimum
no limit: purge grid 0.5–40 %
16.9 %
no limit, purge grid extended to 10−4 %
41.3 %
inlet ≤ 10.25 % (highest published design)
10.0 %
inlet ≤ 6.86 % (the cost model's anchor loop)
6.7 %
inlet ≤ 4.92 % (lowest published design)
4.8 %
CO
N2
CH4
published designs
reactor inlet (mol%, mean over the scored entries)